TWENTY PROPERTIES OF A MORPHISM SURVIVE A FLAT FINITELY PRESENTED FINITE ORDER THICKENING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2597-2656 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2658-2850, the complete author proof (191 body lines).

Editorial note. Navigation stubs: 30 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Base-change stability of the listed morphism properties, finite order thickenings, and the permanence of finite presentation; none of the cited results replaces the converse direction, which is carried out in the retained span.

Source reference. Original lemma algebra-lemma-isomorphism-modulo-locally-nilpotent: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-flatness-morphism-thickenings: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2238–2252. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-thicken-property-morphisms: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 270–284. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-thickening-affine-scheme: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 157–160. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-base-change-affine: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-base-change-etale: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the

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retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-base-change-finite-locally-free: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01TM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02NK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-base-change-smooth: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-base-change-surjective: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-base-change-syntomic: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-base-change-universally-injective: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-etale-flat-etale-fibres: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-finite-flat: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the

retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-finite-presentation-permanence: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-integral-universally-closed: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-permanence-finite-type: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01TH: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-smooth-flat-smooth-fibres: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-syntomic-flat-fibres: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-unramified-etale-fibres: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JY: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02YC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma schemes-lemma-quasi-compact-preserved-base-change: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma schemes-lemma-separated-permanence: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 2850–2850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Consider a commutative diagram*

$$\begin{array}{ccc} (X \subset X') & \xrightarrow[(f,f')]{} & (Y \subset Y') \\ & \searrow \quad \swarrow & \\ & (S \subset S') & \end{array}$$

of thickenings. Assume $Y' \to S'$ locally of finite type, $X' \to S'$ flat and locally of finite presentation, $X = S \times_{S'} X'$, and $Y = S \times_{S'} Y'$. Then

(1) *f is flat if and only if f' is flat,*
(2) *f is an isomorphism if and only if f' is an isomorphism,*
(3) *f is an open immersion if and only if f' is an open immersion,*
(4) *f is quasi-compact if and only if f' is quasi-compact,*
(5) *f is universally closed if and only if f' is universally closed,*
(6) *f is (quasi-)separated if and only if f' is (quasi-)separated,*
(7) *f is a monomorphism if and only if f' is a monomorphism,*
(8) *f is surjective if and only if f' is surjective,*
(9) *f is universally injective if and only if f' is universally injective,*
(10) *f is affine if and only if f' is affine,*
(11) *f is locally quasi-finite if and only if f' is locally quasi-finite,*
(12) *f is locally of finite type of relative dimension d if and only if f' is locally of finite type of relative dimension d,*
(13) *f is universally open if and only if f' is universally open,*
(14) *f is syntomic if and only if f' is syntomic,*
(15) *f is smooth if and only if f' is smooth,*
(16) *f is unramified if and only if f' is unramified,*
(17) *f is étale if and only if f' is étale,*
(18) *f is proper if and only if f' is proper,*
(19) *f is finite if and only if f' is finite,*
(20) *f is finite locally free (of rank d) if and only if f' is finite locally free (of rank d), and*
(21) *add more here.*

Proof. The assumptions on X and Y mean that f is the base change of f' by $X \to X'$. The properties $\mathcal{P}$ listed in (1) – (20) above are all stable under base change, hence if f' has property $\mathcal{P}$, then so does f. See Schemes, Lemmas 01JY, schemes-lemma-quasi-compact-preserved-base-change, schemes-lemma-separated-permanence, and 02YC and Morphisms, Lemmas morphisms-lemma-base-change-surjective, morphisms-lemma-base-change-universally-injective, morphisms-lemma-base-change-affine, 01TM, 02NK, morphisms-lemma-base-change-syntomic, morphisms-lemma-base-change-smooth, 02GA, morphisms-lemma-base-change-etale, 01W4, 01WL, and morphisms-lemma-base-change-finite-locally-free.

The interesting direction in each case is therefore to assume that f has the property and deduce that f' has it too. We make a couple of general remarks which we will use without further mention in the arguments below. (I) Let $W' \subset S'$ be an affine open and let $U' \subset X'$ and $V' \subset Y'$ be affine opens lying over W' with $f'(U') \subset V'$. Let $W' = \mathrm{Spec}(R')$ and denote $I \subset R'$ be the ideal defining the closed subscheme $W' \cap S$. Say $U' = \mathrm{Spec}(B')$ and $V' = \mathrm{Spec}(A')$. Then we get a commutative

diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & IB' & \longrightarrow & B' & \longrightarrow & B & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & IA' & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0 \end{array}$$

with exact rows. (II) The morphisms $X \to X'$ and $Y \to Y'$ are universal homeomorphisms. Hence the topology of the maps f and f' (after any base change) is identical. (III) If f is flat, then f' is flat and $Y' \to S'$ is flat at every point in the image of f', see Lemma lemma-flatness-morphism-thickenings.

Ad (1). This is general remark (III).

Ad (2). Assume f is an isomorphism. Choose an affine open $V' \subset Y'$ and set $U' = (f')^{-1}(V')$. Then $V = Y \cap V'$ is affine which implies that $V \cong f^{-1}(V) = U = Y \times_{Y'} U'$ is affine. By Lemma lemma-thickening-affine-scheme we see that U' is affine. Thus we have a diagram as in the general remark (I). By Algebra, Lemma algebra-lemma-isomorphism-modulo-locally-nilpotent we see that $A' \to B'$ is an isomorphism, i.e., $U' \cong V'$. Thus f' is an isomorphism.

Ad (3). Assume f is an open immersion. Then f is an isomorphism of X with an open subscheme $V \subset Y$. Let $V' \subset Y'$ be the open subscheme whose underlying topological space is V. Then f' is a map from X' to V' which is an isomorphism by (2). Hence f' is an open immersion.

Ad (4). Immediate from remark (II). See also Lemma lemma-thicken-property-morphisms for a more general statement.

Ad (5). Immediate from remark (II). See also Lemma lemma-thicken-property-morphisms for a more general statement.

Ad (6). Note that $X \times_Y X = Y \times_{Y'} (X' \times_{Y'} X')$ so that $X' \times_{Y'} X'$ is a thickening of $X \times_Y X$. Hence the topology of the maps $\Delta_{X/Y}$ and $\Delta_{X'/Y'}$ matches and we win. See also Lemma lemma-thicken-property-morphisms for a more general statement.

Ad (7). Assume f is a monomorphism. Consider the diagonal morphism $\Delta_{X'/Y'} : X' \to X' \times_{Y'} X'$. Observe that $X' \times_{Y'} X' \to S'$ is locally of finite type. The base change of $\Delta_{X'/Y'}$ by $S \to S'$ is $\Delta_{X/Y}$ which is an isomorphism by assumption. By (2) we conclude that $\Delta_{X'/Y'}$ is an isomorphism.

Ad (8). This is clear. See also Lemma lemma-thicken-property-morphisms for a more general statement.

Ad (9). Immediate from remark (II). See also Lemma lemma-thicken-property-morphisms for a more general statement.

Ad (10). Assume f is affine. Choose an affine open $V' \subset Y'$ and set $U' = (f')^{-1}(V')$. Then $V = Y \cap V'$ is affine which implies that $U = Y \times_{Y'} U'$ is affine. By Lemma lemma-thickening-affine-scheme we see that U' is affine. Hence f' is affine. See also Lemma lemma-thicken-property-morphisms for a more general statement.

Ad (11). Follows from the fact that f' is locally of finite type (by Morphisms, Lemma morphisms-lemma-permanence-finite-type) and that quasi-finiteness of a morphism of finite type can be checked on fibres, see Morphisms, Lemma 01TH.

Ad (12). Follows from general remark (II) and the fact that f' is locally of finite type (Morphisms, Lemma morphisms-lemma-permanence-finite-type).

Ad (13). Immediate from general remark (II). See also Lemma lemma-thicken-property-morphisms for a more general statement.

Ad (14). Assume f is syntomic. By Morphisms, Lemma morphisms-lemma-finite-presentation-permanence f' is locally of finite presentation. By general remark (III) f' is flat. The fibres of f' are the fibres of f. Hence f' is syntomic by Morphisms, Lemma morphisms-lemma-syntomic-flat-fibres.

Ad (15). Assume f is smooth. By Morphisms, Lemma morphisms-lemma-finite-presentation-permanence f' is locally of finite presentation. By general remark (III) f' is flat. The fibres of f' are the fibres of f. Hence f' is smooth by Morphisms, Lemma morphisms-lemma-smooth-flat-smooth-fibres.

Ad (16). Assume f unramified. By Morphisms, Lemma morphisms-lemma-permanence-finite-type f' is locally of finite type. The fibres of f' are the fibres of f. Hence f' is unramified by Morphisms, Lemma morphisms-lemma-unramified-etale-fibres.

Ad (17). Assume f étale. By Morphisms, Lemma morphisms-lemma-finite-presentation-permanence f' is locally of finite presentation. By general remark (III) f' is flat. The fibres of f' are the fibres of f. Hence f' is étale by Morphisms, Lemma morphisms-lemma-etale-flat-etale-fibres.

Ad (18). This follows from a combination of (6), the fact that f is locally of finite type (Morphisms, Lemma morphisms-lemma-permanence-finite-type), (4), and (5).

Ad (19). Combine (5), (10), Morphisms, Lemma morphisms-lemma-integral-universally-closed, the fact that f is locally of finite type (Morphisms, Lemma morphisms-lemma-permanence-finite-type), and Morphisms, Lemma 01WJ.

Ad (20). Assume f finite locally free. By (19) we see that f' is finite. By general remark (III) f' is flat. By Morphisms, Lemma morphisms-lemma-finite-presentation-permanence f' is locally of finite presentation. Hence f' is finite locally free by Morphisms, Lemma morphisms-lemma-finite-flat. □